Chapter 2

Policies for Economic Development: Industrial, Trade and Technology Policies in Historical Perspective

2.1. Introduction

In the previous chapter, I pointed out that there have been surprisingly few attempts to apply lessons learned from the historical experiences of developed countries to problems of contemporary development. Also, as will become clearer further on, the few references to these historical experiences tend to be full of myths that support the orthodox version of the history of economic policy in the NDCs, which emphasize the benefits of free trade and *laissez-faire* industrial policy. The story, which underlies virtually all recommendations for Washington Consensus-type policies, goes something like the following.[1]

From the eighteenth century onward, the industrial success of *laissez-faire* Britain proved the superiority of free-market and free-trade policies. Through such policies, which unleashed the entrepreneurial energy of the nation, it overtook interventionist France, its main competitor at the time, establishing itself as the supreme world economic power. Britain was then able to play the role of the architect and hegemon of a new 'Liberal' world economic order, particularly once it had abandoned its deplorable agricultural protection (the Corn Laws) and other remnants of old mercantilist protectionist measures in 1846.

In its quest for this Liberal world order, Britain's ultimate weapon was its economic success based on a free-market/free-trade system; this made other countries realize the limitations of their mercantilist policies and start to adopt free (or at least freer) trade from around 1860. However, Britain was also greatly helped in its project by the works of its classical economists such as Adam Smith and David Ricardo, who theoretically proved the superiority of *laissez-faire* policy, in particular free trade. According to Willy de Clercq, the European Commissioner for External Economic Relations during the early days of the Uruguay Round (1985–9):

> Only as a result of the theoretical legitimacy of free trade when measured against widespread mercantilism provided by David Ricardo, John Stuart Mill and David Hume, Adam Smith and others from the Scottish Enlightenment, and as a consequence of the relative stability provided by the UK as the only and relatively benevolent superpower or hegemon during the second half of the nineteenth century, was free trade able to flourish for the first time [in the late nineteenth century].[2]

This Liberal world order, perfected around 1870, was based on: *laissez-faire* industrial policies at home; low barriers to the international flows of goods, capital and labour; and the macroeconomic stability, both nationally and internationally, which was guaranteed by the Gold Standard and the principle of balanced budgets. A period of unprecedented prosperity followed.

Unfortunately, according to this story, things started to go wrong with the onset of the First World War. In response to the ensuing instability of the world economic and political system, countries once again started to erect trade barriers. In 1930, the USA abandoned free trade and enacted the infamous Smoot–Hawley tariff. According to de Clercq, this tariff 'had disastrous effects on international trade and after a while . . . on American economic growth and employment. Nowadays, some economists even believe that the Great Depression was caused primarily by these tariffs'.[3] The likes of Germany and Japan erected high trade barriers and also started creating powerful cartels, which were closely linked with fascism and these countries' external aggression in the following decades.[4] The world free trade system finally ended in 1932, when Britain, hitherto its champion, succumbed to temptation and reintroduced tariffs. The resulting contraction and instability in the world economy, and then the Second World War, destroyed the last remnants of the first Liberal world order.

After the Second World War, so the story goes, some significant progress was made in trade liberalization through the early GATT (General Agreement on Trade and Tariffs) talks. However, *dirigiste* approaches to economic management dominated the policy-making scene until the 1970s in the developed world, and until the early 1980s in developing countries (as well as the Communist world until its collapse in 1989). According to Sachs and Warner, a number of factors contributed to the pursuit of protectionism and interventionism in developing countries.[5] 'Wrong' theories, such as the infant industry argument, the 'big push' theory of Rosensetin-Rodan (1943), and Latin American structuralism, not to speak of various Marxist theories, prevailed. Protectionist policies were also motivated by political requirements, such as the need for nation building and the need to 'buy off' certain interest groups. There were also legacies of wartime control that persisted into peacetime.

Fortunately, it is held, interventionist policies have been largely abandoned across the world since the 1980s with the rise of Neo-Liberalism, which emphasizes the virtues of small government, *laissez-faire* policies and international openness. By the late 1970s economic growth had begun to falter in most countries in the developing world, with the exception of those in East and Southeast Asia, which were already pursuing 'good' policies. This growth failure, which often manifested itself in the economic crises of the early 1980s, exposed the limitations of old-style interventionism and protectionism.

As a result, most developing countries have come to embrace Neo-Liberal policy reform. The most symbolic of these conversions, according to Bhagwati, are: Brazil's embrace of Neo-Liberal doctrine under the presidency of Fernando Henrique Cardoso, a leading Dependency theorist until the 1980s; the entry of traditionally anti-US Mexico into the NAFTA (North American Free Trade Agreement); and the move towards an open, liberal economy by India, once the bastion of protectionism and regulation.[6] The crowning glory of this trend towards liberalization and opening-up was the fall of Communism in 1989, which finally ended the 'historical anomaly' of a closed world trading system that had prevailed in the early postwar years.[7]

When combined with the establishment of new global governance institutions represented by the WTO, these policy changes at the national level have created a new global economic system, which is comparable in its potential prosperity only to the earlier 'golden age' of Liberalism (1870–1914).[8] Renato Ruggiero, the first Director-General of the WTO, argues that thanks to this new world order we now have 'the potential for eradicating global poverty in the early part of the next [21st] century – a utopian notion even a few decades ago, but a real possibility today'.[9]

As we shall see later, this story paints a powerful but fundamentally misleading picture. Indeed, it should be accepted that there are also some senses in which the late nineteenth century can indeed be described as an era of *laissez-faire*.

To begin with, as we can see in table 2.1, there was a period in the late nineteenth century, albeit a brief one, when liberal trade regimes prevailed in large parts of the world economy. Starting in 1846 with the repeal of the Corn Laws, Britain made a decided shift to a unilateral free trade regime (which was accomplished by the 1860s), although this move was based on its then unchallenged economic superiority and was intricately linked with its imperial policy. Between 1860 and 1880, many European countries reduced tariff protection substantially. At the same time, most of the rest of the world was forced to practice free trade through colonialism (see section 2.3.1) and, in the cases of a few nominally 'independent' countries (such as the Latin American countries, China, Thailand (then Siam), Iran (then Persia) and Turkey (then the Ottoman Empire)), unequal treaties (see section 2.3.2). Of course, the obvious exception to this was the USA, which maintained a very high tariff barrier even during this period. However, given that the USA was still a relatively small part of the world economy, it may not be totally unreasonable to say that this is as close to free trade as the world has ever got (or probably ever will).

More importantly, the scope of state intervention before the First World War (and maybe even up to the Second World War) was quite limited by modern standards. For example, before the 1930s, both the hegemony of the doctrine of balanced budget and the limited scope for taxation (given, among other things, the absence of personal and corporate income taxes in most countries) severely limited the scope for active budgetary policy. The narrow tax base restricted government budgets, so large fiscal outlays for developmental purposes were difficult, even if the government had the intention to make them – railways being an obvious exception in a number of countries. In most countries, fully-fledged central banking did not exist until the early twentieth century, so the scope for monetary policy was also very limited. On the whole, banks were privately-owned and little regulated by the state, so the scope for using 'directed credit programmes', which were so widely and successfully used in countries like Japan, Korea, Taiwan and France during the postwar period, was extremely limited. Measures like the nationalization of industry and indicative investment planning, practices that served many European countries, especially France, Austria and Norway, well in the early postwar years, were regarded as unthinkable outside wartime before the Second World War. One somewhat paradoxical consequence of all these limitations was that tariff protection was far more important as a policy tool in the nineteenth century than it is in our time.

Table 2.1

Average Tariff Rates on Manufactured Products for Selected Developed Countries in Their Early Stages of Development[1]
(weighted average; in percentages of value)

	1820[2]	1875[2]	1913	1925	1931	1950
Austria[3]	R	15–20	18	16	24	18
Belgium[4]	6–8	9–10	9	15	14	11
Denmark	25–35	15–20	14	10	n.a.	3
France	R	12–15	20	21	30	18
Germany[5]	8–12	4–6	13	20	21	26
Italy	n.a.	8–10	18	22	46	25
Japan[6]	R	5	30	n.a.	n.a.	n.a.
Netherlands[4]	6–8	3–5	4	6	n.a.	11
Russia	R	15–20	84	R	R	R
Spain	R	15–20	41	41	63	n.a.
Sweden	R	3–5	20	16	21	9
Switzerland	8–12	4–6	9	14	19	n.a.
United Kingdom	45–55	0	0	5	n.a.	23
United States	35–45	40–50	44	37	48	14

Source: Bairoch 1993, p. 40, table 3.3.

Notes:

R = Numerous and important restrictions on manufactured imports existed and therefore average tariff rates are not meaningful.

1. World Bank (1991, p. 97, Box table 5.2) provides a similar table, partly drawing on Bairoch's own studies that form the basis of the above table. However, the World Bank figures, although in most cases very similar to Bairoch's figures, are ***unweighted*** averages, which are obviously less preferable to the ***weighted*** average figures that Bairoch provides.
2. These are very approximate rates, and give range of average rates, not extremes.
3. Austria–Hungary before 1925.
4. In 1820, Belgium was united with the Netherlands.
5. The 1820 figure is for Prussia only.
6. Before 1911, Japan was obliged to keep low tariff rates (up to 5%) through a series of 'unequal treaties' with the European countries and the USA. The World Bank table cited in note 1 above gives Japan's ***unweighted*** average tariff rate for ***all goods*** (not just manufactured goods) for the years 1925, 1930, 1950 as 13%, 19%, 4%.

Despite these limitations, as I have pointed out in Chapter 1 and will show in more detail in the rest of this chapter, virtually all NDCs actively used interventionist industrial, trade and technology (ITT) policies that are aimed at promoting infant industries during their catch-up periods.[10] As we shall see later, there were some apparent exceptions to this, such as Switzerland and the Netherlands, but these were countries that were either at or very near the technological frontier and thus did not, by definition, need much infant industry promotion. Some countries used activist ITT policies even after the catch-up was successfully achieved (Britain in the early nineteenth century, the USA in the early twentieth century). Tariff protection was obviously a very important policy tool in the ITT policy package used by the NDCs, but, as we shall show later, it was by no means the only one used, or even necessarily the most important.

On the trade front, subsidies and duty drawbacks on inputs for exported goods were frequently used to promote exports. Governments both provided industrial subsidies and used various public investment programmes, especially in infrastructure but also in manufacturing. They supported foreign technology acquisition, sometimes by legal means such as financing study tours and apprenticeships, and sometimes through illegal measures, which included support for industrial espionage, smuggling of contraband machinery and refusal to acknowledge foreign patents. Development of domestic technological capabilities was encouraged through financial support for research and development, education and training. Measures were also taken to raise awareness of advanced technologies (for example, the establishment of model factories, organisation of exhibitions, granting of free imported machinery to private sector firms). In addition, some governments created institutional mechanisms that facilitated public–private cooperation (for example, public–private joint ventures and industry associations with close links with the government). It is important to note that many of these policies are greatly frowned upon these days, even when they have not been made explicitly illegal through bilateral and multilateral agreements.

When they reached the technological frontier, the NDCs used a range of policies in order to help themselves pull away from their existing and potential competitors. Britain, given the duration for which it held the position of 'frontier economy', is most visible in this respect, but other countries also used similar measures when they could. Britain used measures to control transfer of technology to its potential competitors (for example, controls on skilled worker migration or machinery export), and put pressure on the less developed countries to open up their markets, by force if necessary. However, the catch-up economies that were not formal or informal colonies did not simply sit down and accept these restrictive measures. They employed a wide variety of measures to overcome the obstacles created by these restrictions, even resorting to 'illegal' means, such as the poaching of workers and smuggling of machinery.[11]

2.2. The Catch-up Strategies

In this section, I examine the experiences of a range of NDCs – Britain, the USA, Germany, France, Sweden, Belgium, the Netherlands, Switzerland, Japan, Korea and Taiwan – and consider what kinds of industrial, trade and technology (ITT) policies they used when they themselves were developing countries. I show that in most of these countries, the policies that were used are almost the opposite of what the present orthodoxy says they employed 'and currently recommends that the currently developing countries should also use'.

2.2.1. Britain

As the intellectual fountain of the modern *laissez-faire* doctrines, and as the only country that can claim to have practised a total free trade at one stage in its history, Britain is widely regarded as having developed without significant state intervention. However, this could not be further from the truth.

Britain entered its post-feudal age (thirteenth and fourteenth centuries) as a relatively backward economy. Before 1600, it was an importer of technology from the Continent.[12] It relied on exports of raw wool and, to a lesser extent, of low-value-added wool cloth (what was then known as 'short cloth') to the then more advanced Low Countries, especially the towns of Bruges, Ghent and Ypres in Flanders, now part of Belgium.[13] The British monarchs of this time taxed these products mainly for revenue reasons, but since cloth was taxed more lightly than raw wool, this encouraged import substitution in wool cloth and a certain amount of export success.[14] Edward III (1327–77) is believed to have been the first king who deliberately tried to develop local wool cloth manufacturing. He wore only English cloth to set an example to the rest of the country,[15] brought in Flemish weavers, centralized trade in raw wool and banned the import of woollen cloth.[16]

The Tudor monarchs gave further impetus to the development of this industry with what can only be described as a deliberate infant industry promotion policy. The celebrated eighteenth-century merchant, politician and novelist, Daniel Defoe, describes this policy in his now-almost-forgotten book, *A Plan of the English Commerce* (1728).[17] In it, he describes in some detail how the Tudor monarchs, especially Henry VII (1485–1509) and Elizabeth I (1558–1603), transformed England from a country relying heavily on raw wool export to the Low Countries into the most formidable wool-manufacturing nation in the world.[18]

According to Defoe, Henry VII had, prior to his coronation in 1485, 'been a kind of a Refugee in the Court of his Aunt the Dutchess of *Burgundy* [italics original]'.[19] There, he was deeply impressed by the prosperity in the Low Countries based on wool manufacturing, and from 1489 onwards he put in place schemes to promote British wool manufacturing. The measures used included sending royal missions to identify locations suited to wool manufacturing,[20] poaching skilled workers from the Low Countries,[21] increasing duties on, and even temporarily banning the export of, raw wool. Ramsay also documents the legislation in 1489, 1512, 1513 and 1536, which banned the exports of unfinished cloths, save for coarse pieces below a certain market value. This, he observes, reflected the then 'influential view that if it was preferable to export wool in the form of cloth rather than in the raw state then it was likewise better to ship cloth fully dressed and dyed than in a semi-manufactured state, "unbarbed and unshorn" '.[22]

As Defoe emphasizes, Henry VII realized that, given Britain's technology gap with the Low Countries, this transformation was going to take a long time, and therefore he took a gradualist approach.[23] Therefore, he raised export duties on raw wool only when the industry was better established. As soon as it became clear that Britain simply did not have the capacity to process all the raw wool it produced, he withdrew the ban on raw wool export he had imposed.[24] According to Defoe, it was not until the time of Elizabeth I (1587), nearly a hundred years after Henry VII started his import substitution policy (1489), that Britain was confident enough about its wool manufacturing industry's international competitiveness to ban raw wool export completely.[25] This eventually drove the manufacturers in the Low Countries to ruin.

According to Defoe's analysis, other factors besides this import substitution policy helped the achievement of British victory in the wool industry under Elizabeth I. Some of these factors were fortuitous, such as the migration of Protestant Flemish textile workers following the war of independence from Spain in 1567. However, other elements were deliberately created by the state. In order to open new markets, Elizabeth I dispatched trade envoys to the Pope and the Emperors of Russia, Mogul, and Persia. Britain's massive investment in building its naval supremacy allowed it to break into new markets and often to colonise them and keep them as captive markets.[26]

It is difficult to establish the relative importance of the above-mentioned factors in explaining the British success in wool manufacturing. However, it does seem clear that, without what can only be described as the sixteenth-century equivalent of modern infant industry promotion strategy put in place by Henry VII and further pursued by his successors, it would have been very difficult, if not necessarily impossible, for Britain to achieve this initial success in industrialization: without this key industry, which accounted for at least half of Britain's export revenue during the eighteenth century, its Industrial Revolution might have been very difficult, to say the least.[27]

The 1721 reform of the mercantile law introduced by Robert Walpole, the first British Prime Minister, during the reign of George I (1714–27) signified a dramatic shift in the focus of British industrial and trade policies.

Prior to this, the British government's policies were in general aimed at capturing trade (most importantly through colonialization and the Navigation Acts, which required that trade with Britain had to be conducted in British ships[28]) and at generating government revenue. The promotion of wool manufacturing, as discussed above, was the most important exception to this, but even this was partly motivated by the desire to generate more government revenue. In contrast, the policies introduced after 1721 were deliberately aimed at promoting manufacturing industries. Introducing the new law, Walpole stated, through the king's address to Parliament: 'it is evident that nothing so much contributes to promote the public well-being as the exportation of manufactured goods and the importation of foreign raw material'.[29]

The 1721 legislation, and its subsequent supplementary policy changes, included the following measures.[30] First of all, import duties on raw materials used for manufactures were lowered, or even dropped altogether.[31] Second, duty drawbacks on imported raw materials for exported manufactures – a policy that had been well established in the country since the days of William and Mary – were increased.[32] For example, the duty on beaver skins was reduced and in case of export a drawback of half the duty paid was allowed.[33] Third, export duties on most manufactures were abolished.[34] Fourth, duties on imported foreign manufactured goods were significantly raised. Fifth, export subsidies ('bounties') were extended to new items like silk products (1722) and gunpowder (1731), while the existing export subsidies to sailcloth and refined sugar were increased (in 1731 and 1733 respectively).[35] Sixth, regulation was introduced to control the quality of manufactured products, especially textile products, so that unscrupulous manufacturers could not damage the reputation of British products in foreign markets.[36]

Brisco sums up the principle behind this new legislation as follows: '[manufacturers] had to be protected at home from competition with foreign finished products; free exportation of finished articles had to be secured; and where possible, encouragement had to be given by bounty and allowance'.[37] What is very interesting to note here is that the policies introduced by the 1721 reform, as well as the principles behind them, were uncannily similar to those used by countries like Japan, Korea and Taiwan during the postwar period, as we shall see below (section 2.2.7).

With the Industrial Revolution in the second half of the eighteenth century, Britain started widening its technological lead over other countries. However, even then it continued its policy of industrial promotion until the mid-nineteenth century, by which time its technological supremacy was overwhelming.[38]

The first and most important component of this was clearly tariff protection. As we saw from table 2.1, Britain had very high tariffs on manufacturing products as late as the 1820s, some two generations after the start of its Industrial Revolution, and when it was significantly ahead of its competitor nations in technological terms. Measures other than tariff protection were also deployed.

First of all, Britain banned the imports of superior products from some of its colonies if they happened to threaten British industries. In 1699, the Wool Act prohibited exports of woollen products from the colonies, killing off the then superior Irish wool industry (see section 2.3). In 1700, a ban was imposed on the imports of superior Indian cotton products ('calicoes'), debilitating what was then arguably the world's most efficient cotton manufacturing sector. The Indian cotton industry was subsequently destroyed by the ending of the East India Company's monopoly in international trade in 1813, when Britain had become a more efficient producer than India (see section 2.3). By 1873, two generations after the event, it was already estimated that 40–45% of all British cotton textile exports went to India.[39]

By the end of the Napoleonic Wars in 1815, however, there were increasing pressures for free trade in Britain from the increasingly confident manufacturers. By this time, most British manufacturers were firmly established as the most efficient in the world in most industries, except in a few limited areas where countries like Belgium and Switzerland possessed technological leads over Britain (see section 2.2.6). Although a new Corn Law passed in 1815 (Britain had had numerous Corn Laws dating back to 1463) meant an increase in agricultural protection, the pressure for freer trade was building up.[40]

Although there was a round of tariff reduction in 1833, the big change came in 1846, when the Corn Law was repealed and tariffs on many manufacturing goods abolished.[41] The repeal of the Corn Law is these days commonly regarded as the ultimate victory of the Classical Liberal economic doctrine over wrong-headed mercantilism. Although we should not underestimate the role of economic theory in this policy shift, many historians more familiar with the period point out that it should probably be understood as an act of 'free trade imperialism'[42] intended to 'halt the move to industrialisation on the Continent by enlarging the market for agricultural produce and primary materials'.[43]

Indeed, many key leaders of the campaign to repeal the Corn Law, such as the politician Robert Cobden and John Bowring of the Board of Trade, saw their campaign in precisely such terms.[44] Cobden's view on this is clearly revealed in the following passage:

> The factory system would, in all probability, not have taken place in America and Germany. It most certainly could not have flourished, as it has done, both in these states, and in France, Belgium, and Switzerland, through the fostering bounties which the high-priced food of the British artisan has offered to the cheaper fed manufacturer of those countries'.[45]

Symbolic though the repeal of Corn Law may have been, the real shift to free trade only happened in the 1850s. It was only after Gladstone's budgets of the 1850s, and especially that of 1860, in conjunction with the Anglo-French free trade treaty (the so-called Cobden–Chevalier Treaty) signed that year, that most tariffs were eliminated. The following passage succinctly describes the magnitude of trade liberalization that happened in Britain during the 1850s. 'In 1848, Britain had 1,146 dutiable articles; by 1860 she had forty-eight, all but twelve being revenue duties on luxuries or semi-luxuries. Once the most complex in Europe, the British tariff could now be printed "on half a page of Whitaker's Almanack" '.[46]

It is important to note here that Britain's technological lead that enabled this shift to a free trade regime had been achieved 'behind high and long-lasting tariff barriers'.[47] It is also important to note that the overall liberalization of the British economy that occurred during the mid-nineteenth century, of which trade liberalization was just a part, was a highly controlled affair overseen by the state, and not achieved through a *laissez-faire* approach.[48] It should also be pointed out that Britain 'adopted Free Trade painfully slowly: eighty-four years from *The Wealth of Nations* to Gladstone's 1860 budget; thirty-one from Waterloo to the ritual victory of 1846'.[49]

Moreover, the free-trade regime did not last long. By the 1880s, some hard-pressed British manufacturers were asking for protection. By the early twentieth century, reintroduction of protectionism was one of the hottest issues in British politics, as the country was rapidly losing its manufacturing advantage to the USA and Germany: testimony to this was the influence of the Tariff Reform League, formed in 1903 under the leadership of the charismatic politician Joseph Chamberlain.[50] The era of free trade ended when Britain finally acknowledged that it had lost its manufacturing eminence and re-introduced tariffs on a large scale in 1932.[51]

2.2.2. USA

As List pointed out (see Chapter 1), Britain was the first country successfully to launch an infant industry promotion strategy. However, its most ardent user was probably the USA – the eminent economic historian Paul Bairoch once called it 'the mother country and bastion of modern protectionism'.[52]

This fact is rarely acknowledged in the modern literature, however, especially that coming out of the USA, and even many otherwise knowledgeable people do not seem to be aware of it. No less an economic historian than Clive Trebilcock, an authority on European Industrial Revolution, when commenting on the introduction of 1879 tariffs in Germany, stated that tariffs were going up all over the world, including 'even free-trade America'.[53]

Even when the existence of high tariffs is acknowledged, their importance is severely downplayed. For example, in what was until recently the standard overview of US economic history, North mentions tariffs once, only to dismiss them as an insignificant factor in explaining US industrial development. He argues, without bothering to establish the case and by citing only one highly-biased secondary source (the classic study by F Taussig, 1892), 'while tariffs became increasingly protective in the years after the Civil War, it is doubtful if they were very influential in affecting seriously the spread of manufacturing'.[54]

However, a more careful and unbiased reading of the history reveals that the importance of infant industry protection in US development cannot be overemphasized. From the early days of colonization in what later became the USA, protection of domestic industry was a controversial policy issue. To begin with, Britain did not want to industrialize the colonies and duly implemented policies to that effect (see section 2.3. for further details). Around the time of independence, Southern agrarian interests opposed any protection, while Northern manufacturing interests – represented by, among others, Alexander Hamilton, the first Secretary of the Treasury of the USA (1789–95) – wanted it.[55]

Indeed, many point out that it was Alexander Hamilton, in his *Reports of the Secretary of the Treasury on the Subject of Manufactures* (1791), and not Friedrich List as is often thought, who first systematically set out the infant industry argument.[56] In fact, as Henderson and Reinert point out, List started out as a free trade advocate and only converted to the infant industry argument following his period of exile in the USA (1825–30). While he was there, he was exposed to the works of Alexander Hamilton and the then leading US economist and strong advocate of infant industry protection, Daniel Raymond.[57]

In his *Reports*, Hamilton argued that competition from abroad and 'forces of habit' would mean that new industries that could soon become internationally competitive ('infant industries')[58] would not be started in the USA, unless their initial losses were guaranteed by government aid. This aid, he said, could take the form of import duties or, in rare cases, prohibition of import.[59] It is interesting to note that there is a close resemblance between this view and that espoused by Walpole (see section 2.2.1) – a point that was not lost on the contemporary Americans, especially Hamilton's political opponents.[60] In turn, it should also be noted that both the Walpolean and the Hamiltonian views are remarkably similar to the view that lies behind East Asia's postwar industrial policy (see section 2.2.7.).

Initially, the USA did not have a federal-level tariff system, and an attempt to grant the Congress tariff power in 1781 failed.[61] When it acquired the power to tax, the Congress passed a liberal tariff act (1789), imposing a five per cent flat rate tariff on all imports, with some exceptions, such as hemp, glass, and nails. Many tariffs were increased in 1792, although they still fell far short of Hamilton's recommendations, which called for an extensive system of infant industry protection and subsidies. After that, until the war with Britain in 1812, the average tariff level remained around 12.5 per cent, but in order to meet the increased wartime expenses, all tariffs were doubled in 1812.[62]

A significant shift in policy occurred in 1816, when, as List noted (Chapter 1), a new law was introduced to keep the tariff level close to that from wartime as a result of the considerable political influence of the infant industries that had grown up under the 'natural' protection accorded by the war with Britain. This was done despite the fact that the revenue was no longer needed – especially protected were cotton, woollen, and iron goods.[63] In the 1816 tariff law, almost all manufactured goods were subject to tariffs of around 35 per cent.[64] Table 2.1 shows that the average tariff level for manufacturing products in the USA in 1820 was around 40 per cent. Initially, this measure was welcomed by everyone, including the Southern states, which hoped that it would help industries to grow in their territories. However, the Southern states soon turned against it because of their interests in importing superior quality British manufactures and because of the failure of the industries to emerge in their own territories.[65]

The Southern agrarian interests, with the help of the New England (and especially New York) shippers, were able to defeat bills calling for higher tariffs in 1820, 1821 and 1823.[66] However, in 1824, a new, still higher, tariff was enacted. In 1828 the so-called Tariff of Abominations further divided the country. This was because this time the northern and western agricultural interests were adding high tariffs on the raw materials or low value-added manufactures that they produced (e.g., wool, hemp, flax, fur and liquor), thus creating tension with the New England manufacturing states.[67]

Yet another tariff law was passed in 1832. This offered a 40 per cent tariff rate on average for manufactured goods – a much lower cut than the Southerners had wanted – and particularly high protection was accorded to iron and textile goods (e.g., 40–45 per cent on woollen manufactured goods and 50 per cent for clothing). This led to the so-called Nullification Crisis, started by South Carolina's refusal to accept the law. A compromise bill was passed in 1833, which offered few immediate reductions but made a provision for gradual reduction over the next ten years, down to about 25 per cent for manufactured goods and 20 per cent for all goods. However, as soon as this ten-year reduction ended in 1842, a new tariff act was passed, raising duties back up to about the 1832 levels.[68]

There was a reduction in protection in the 1846 tariff law, although the average *ad valorem* duty on the 51 most important categories of imported goods was still 27 per cent. There was a further reduction in 1857, made possible by the coalition of the Democrats, the cloth manufacturers who wanted raw wool placed on the 'free list', and railroad interests who wanted tariff-free iron from abroad. Bairoch describes the period between 1846 and 1861 as one of 'modest protectionism'.[69] However, this protectionism is only 'modest' by the historical standards of the USA (see table 2.1). It must also be pointed out that, given the high transportation costs of the period, which prevailed at least until the 1870s, US tariffs would have been a greater barrier to international trade than the European ones, even if both had been at the same level.[70]

However, the tension surrounding both the tariff and slave issues persisted between North and South, and finally culminated in the Civil War (1861–5). The Civil War is commonly thought to have been fought solely over the issue of slavery, but in fact tariffs were another important issue. Garraty and Carnes state that '[a] war against slavery would not have been supported by a majority of Northerners. Slavery was the root case of secession but not of the North's determination to resist secession, which resulted from the people's commitment to the union'.[71] Given that the South had seen tariffs as the major existing liability of the union while the abolition of slavery was still only a theoretical possibility, the importance of the tariff issue in causing the secession cannot be over-emphasized.

Lincoln's victory in the presidential election of 1860 would have been very difficult, if not impossible, had the leading protectionist states of Pennsylvania and New Jersey not switched their allegiance to the Republican Party thanks to its election pledge to maintain increased protection.[72] The pledge (the 'twelfth plank' of the platform) was deliberately worded ambiguously in such a way as to assuage the free-trade element in the Party.[73] At the same time, it was still acceptable to the protectionist states, given that Lincoln was a 'true blue protectionist' and thus seen as someone who would live up to the spirit of the pledge, once elected.[74]

Early in his political career Lincoln had been a leading member of the hard-line protectionist Whig party and an enthusiastic follower of the charismatic politician, Henry Clay. Clay advocated the 'American

System', which consisted of infant industry protection ('Protection for Home Industries') and infrastructural development ('Internal Improvements'), in explicit opposition to the 'British System' of free trade, and Lincoln fully shared his view.[75] Although during the campaign Lincoln was compelled to keep quiet on most of the controversial issues, including tariffs, in order to hold a diverse and young party together[76], he unwaveringly gave assurance of his protectionist belief when it was deemed necessary.[77]

Although he was consistently anti-slavery, Lincoln had never before advocated forceful abolition of slavery; he considered blacks racially inferior, and was against black suffrage. Given this, there was probably less to fear for the South on the slavery front than on the tariff front upon his election. Indeed, even in the early days of the Civil War, Lincoln made it clear that he was quite willing to allow slavery in the Southern states in order to keep the Union together. He enacted slave emancipation in the autumn of 1862 as a strategic move to win the war rather than out of moral conviction.[78]

In 1862, a new tariff act was introduced. It was disguised as 'compensation' for the increased excise tax and the emergency income tax during the Civil War, in order that the previous margin of protection could be maintained. This raised the rates to 'their highest level in thirty years – much higher, in many cases, than the new excise taxes warranted'.[79] In 1864, tariffs were raised still further, to their highest ever rates, to meet the demands of war expenditure; they remained at those levels after the war, although other internal taxes were repealed.[80] In this way, the victory of the North in the Civil War ensured that the USA remained the most ardent practitioner of infant industry protection until the First World War, and even until the Second World War – with the notable exception of Russia in the early twentieth century (see Table 2.1).[81]

In 1913, following the previous Democratic electoral victory, the Underwood Tariff bill was passed, leading to 'a large increase in the categories of goods allowed free entry and to a substantial drop in average import duties';[82] this reduced the average tariff on manufactured goods from 44 per cent to 25 per cent. However, the onset of the First World War made this bill ineffective and a new emergency tariff legislation was put in place by 1922, following the Republican return to power in 1921. In the 1922 law, although the tariffs did not return to their high 1861–1913 levels, the percentage effectively paid on manufactured imports rose by 30 per cent.[83]

Following the start of the Great Depression there came the 1930 Smoot-Hawley tariff – 'the most visible and dramatic act of anti-trade folly', according to Bhagwati.[84] However, this characterization is very misleading. While the Smoot-Hawley tariff provoked an international tariff war, thanks to its bad timing – especially given the new status of the USA as the world's largest creditor nation following the First World War – it did not constitute a radical departure from the country's traditional trade policy stance.[85]

In fact, the Smoot-Hawley tariff only marginally increased the degree of protectionism in the US economy. As we can see from table 2.1, the average tariff rate for manufactured goods that resulted from this bill was 48 per cent, which still falls within the range of the average rates that had prevailed in the USA since the Civil War, albeit in the upper region of this range. It is only in relation to the brief 'Liberal' interlude of 1913–29 that the 1930 tariff bill can be interpreted as increasing protectionism, although even then it was not by very much. Table 2.1 shows that the average rate of tariff on manufactures in 1925 was 37 per cent and rose to 48 per cent in 1931.

It was only after the Second World War that the USA – with its industrial supremacy unchallenged – finally liberalized its trade and started championing the cause of free trade. However, it should be noted that the USA never practised free trade to the same degree as Britain did during its free-trade period (1860 to 1932). It never had a zero-tariff regime like that of the UK, and it was much more aggressive in using 'hidden' protectionist measures. These included: VERs (voluntary export restraints); quotas on textile and clothing (through the Multi-Fibre Agreement); protection and subsidies for agriculture (compare this with the repeal of the Corn Laws in Britain); and unilateral trade sanctions (especially through the use of anti-dumping duties).[86]

In contrast to the attitude of a generation ago, represented by the above-mentioned work of North, there is now a growing recognition of the importance of protectionism among US economic historians, who used to be extremely wary of saying anything positive about it. Today, there seems at least to be a consensus that tariff protection was critical in the development of certain key industries, such as the textile industry in the early nineteenth century and the iron and steel industries in the second half of the nineteenth century.[87] Although some commentators doubt whether the overall national welfare effect of protectionism was positive, the US growth records during the protectionist period make this scepticism look overly cautious, if not downright biased.

Bairoch points out that, throughout the nineteenth century and up to the 1920s, the USA was the fastest growing economy in the world, despite being the most protectionist during almost all of this period.[88] There is also no evidence that the only significant reduction of protectionism in the US economy, between 1846 and 1861, had any noticeable positive impact on the country's development. Most interestingly, the two best 20-year GDP per capita growth performances during the 1830–1910 period were 1870–1890 (2.1 per cent) and 1890–1910 (two per cent) – both periods of particularly high protectionism.[89] It is hard to believe that this association between the degree of protectionism and overall growth is purely coincidental. Indeed, O'Rourke shows some statistical evidence from ten NDCs, including the USA, during the 'liberal golden age' of 1875–1914, to the effect that protection (measured by average tariff rates) was positively related to growth.[90]

Of course, as many people point out, tariff protection for some industries certainly outlived its usefulness. For example, despite the continuing debate on this issue,[91] it is widely agreed that by the 1830s, American cotton textile producers would not have needed protection, particularly in certain low-value-added segments of the market.[92] It is also very likely that even some of the necessary tariffs may have been set at excessively high levels due to interest-group pressures and the complicated horse-trading that has characterised the country's policy making. Despite these qualifications it seems difficult to deny that, without infant industry protection, the US economy would not have industrialized and developed as fast as it did in its catching-up period.

Important as it may have been, tariff protection was not the only policy deployed by the US government in order to promote the country's economic development during its catch-up phase. From the Morrill Act of 1862, and probably from as early as the 1830s, the government supported an extensive range of agricultural research. Measures used included the granting of government land to agricultural colleges and the establishment of government research institutes, such as the Bureau of Animal Industry and the Bureau of Agricultural Chemistry. In the second half of the nineteenth century, it expanded public educational investments – in 1840, less than half of the total investment in education was public, whereas by 1900 this figure had risen to almost 80 per cent – and raised the literacy ratio to 94 per cent by 1900. The role of the US government in promoting the development of transportation infrastructure, especially through the granting of land and subsidies to railway companies, was also critical in shaping the country's developmental path.[93]

It is important to recognize that the role of the US federal government in industrial development has been substantial even in the postwar era, thanks to the large amount of defence-related procurements and R&D spending, which have had enormous spillover effects.[94] The share of the US federal government in total R&D spending, which was only 16 per cent in 1930,[95] remained between one-half and two-thirds during the postwar years.[96] Industries such as computers, aerospace and the internet, where the USA still maintains an international edge despite the decline in its overall technological leadership, would not have existed without defence-related R&D funding by the country's federal government.[97] The critical role of the US government's National Institutes of Health (NIH) in supporting R&D in pharmaceutical and biotechnology industries, thus maintaining the US lead in these industries, should also be mentioned. Even according to the information provided by the US pharmaceutical industry association, only 43 per cent of pharmaceutical R&D is funded by the industry itself, while 29 per cent is funded by the NIH.[98]

During the nineteenth century, the USA was not only the strongest bastion of protectionist policies, but was also their intellectual home. At that time it was widely believed among US intellectuals that 'the new country required a new economics, one grounded in different political institutions and economic conditions than those prevailing in the Old World'.[99] Some of them went so far as to argue that even internationally competitive US industries should have tariff protection because of the possibility of predatory dumping by large European enterprises, who, after decimating the American firms, would revert to monopolistic pricing.[100]

Well into the last quarter of the nineteenth century, most of the more original US economists of the period seem to have been strong advocates of infant industry protection. The well-known supporters of infant industry promotion, Daniel Raymond (who influenced Friedrich List) and Mathew Carey were the two leading economists of the early nineteenth century, while American economics during the mid-to late nineteenth century was dominated by Carey's son Henry. Henry Carey was described as 'the only American economist of importance' by Marx and Engels in the early 1850s[101] and was one of Lincoln's (somewhat frustrated) economic advisors.[102] Unfortunately, most of these economists have now been airbrushed out of the history of US economic thought, but it was they, rather than the American Classical economists (then regarded as second-rate by the British standard), who were the more prominent intellectual figures of the time.

What is especially interesting to note here is that many US intellectuals and politicians during the country's catch-up period clearly understood that the free trade theory advocated by the British Classical Economists was unsuited to their country. Reinert reports that, due to this concern, Thomas Jefferson tried (in vain) to prevent the publication of Ricardo's *Principles* in the USA.[103] Reinert also cites from List's work the comment by a US Congressman, a contemporary of List, who observed that English trade theory 'like most English manufactured goods, is intended for export, not for consumption at home'.[104]

As I mentioned earlier, Henry Clay, the most prominent protectionist politician of the early nineteenth century and Abraham Lincoln's early mentor, named his economic policy platform the 'American System', in explicit opposition to what he called the 'British System' of free trade. Somewhat later, Henry Carey even argued that free trade was part of British imperialist system that consigned the USA to a role of primary product exporter.[105] It is also reported that during the 1860 election campaign, in which Carey played a key intellectual role, the Republicans in some protectionist states referred disparagingly to the Democrats as a 'Southern-*British*-Antitariff-Disunion party [italics added]'.[106]

2.2.3. Germany

Germany is now commonly known as the home of infant industry protection, both intellectually and in terms of policies. However, historically speaking, tariff protection actually played a far less important role in the economic development of Germany than in that of the UK or the USA.

The tariff protection for industry in Prussia before the 1834 German customs union under its leadership (*Zollverein*) and that subsequently accorded to German industry in general remained mild. Trebilcock, an authority on German industrialization of the period, categorically states that '*Zollverein* tariffs were insufficient to provide effective "infant industry" protection; even the iron manufacturers went without tariff duties until 1844, and lacked successful protection well beyond this'.[107] The Prussian state constantly resisted political pressures for higher tariffs by other member states of *Zollverein*. Although there were increases in tariffs in 1844 (on iron) and 1846 (on cotton yarn), these were relatively